Project 3 Report: Designing and Conducting an A/B Test

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1. INTRODUCTION

This project aims to evaluate how interface layout affects user behavior in a data analysis application. Specifically, we conduct an A/B test comparing two versions of a Shiny app that differ only in the placement of the control panel.

In the left-panel version, the control panel is located on the left side of the interface, while in the right-panel version, the control panel is placed on the right side. The goal is to determine whether one layout leads to improved user engagement and task performance.

We hypothesize that the layout of the control panel may influence how users interact with the application, affecting metrics such as click-through rate, time spent, task completion, and bounce rate.

2. EXPERIMENTAL DESIGN

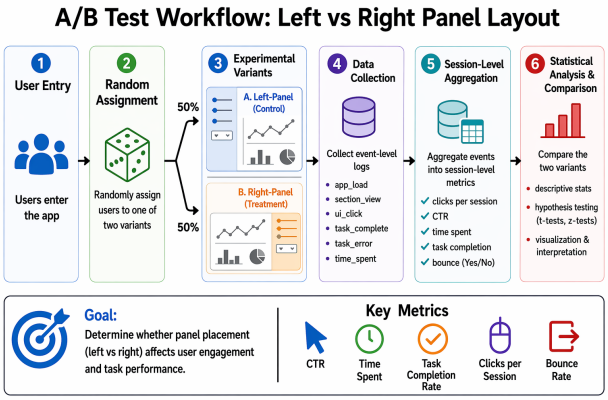

Figure 1. A/B testing workflow showing user assignment and data collection process

As illustrated in Figure 1, users are randomly assigned to either the left-panel or right-panel layout, and their interactions are tracked for analysis.

We designed a controlled A/B experiment with two variants: a left-panel layout (control) and a right-panel layout (treatment). Users were randomly assigned at the user level in a 1:1 ratio, and each user was exposed to only one variant throughout the experiment.

While the assignment was balanced at the user level, the number of sessions is not necessarily equal across variants, as individual users may generate different numbers of sessions.

Both versions of the app were identical in functionality and content, differing only in the placement of the control panel. This ensures that any observed differences in user behavior can be attributed to the layout change.

Key performance metrics include:

- Click-through rate (CTR)
- Number of clicks per session
- Total time spent in the application
- Task completion rate
- Bounce rate

The analysis is conducted at the session level to capture overall user interaction patterns.

3. DATA COLLECTION

The dataset consists of event-level logs generated in a simulated environment designed to mimic realistic user interactions. Each row represents a user event, such as page views, button clicks, task completion, or time spent in different sections.

The synthetic data were constructed to capture a range of plausible user behaviors, including short sessions, highly engaged sessions, click interactions, task completions, drop-offs, and occasional task errors. This design aims to approximate real-world usage patterns within a controlled experimental setting.

The dataset includes approximately 95 sessions, with 44 sessions in the left-panel variant and 51 sessions in the right-panel variant.

To enable meaningful analysis, the event-level data were aggregated into a session-level dataset. For each session, we computed key metrics including the number of clicks, number of section views, total time spent, whether any task was completed, and whether the session bounced.

Although the dataset is relatively small and simulated, it provides a reasonable approximation of user interaction patterns for the purpose of demonstrating the A/B testing workflow and analysis.

4. STATISTICAL ANALYSIS AND RESULTS

We first summarize the key metrics for each variant. The results show that the right-panel layout consistently outperforms the left-panel layout across multiple measures.

Table 1. Summary Statistics by Variant

Metric	Left	Right
Clicks per session	3.386	4.235
CTR	0.697	0.864
Avg time (sec)	99.341	112.157
Completion rate	0.568	0.647
Bounce rate	0.227	0.098

Specifically, the right-panel variant shows higher average clicks per session (4.235 vs 3.386), higher click-through rate (0.864 vs 0.697), and longer average time spent (112.157 vs 99.341 seconds). It also achieves a higher task completion rate (0.647 vs 0.568) and a lower bounce rate (0.098 vs 0.227).

In terms of effect size, this corresponds to an absolute increase of 0.849 clicks per session (approximately 25.1% relative lift), a CTR increase of 0.167 (about 24.0% relative lift), and a task completion improvement of 0.079 (around 13.9%). The bounce rate decreases by 0.129, indicating improved user retention.

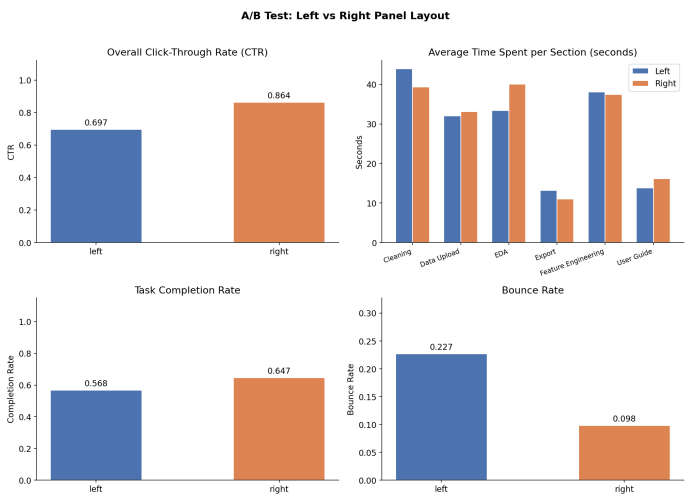

Figure 2. Comparison of CTR, time spent, task completion rate, and bounce rate between left and right panel layouts.

4.1. Statistical Tests

To evaluate whether the observed differences are statistically significant, we conducted two-sample t-tests for continuous variables and proportion z-tests for binary outcomes.

The results indicate that none of the differences are statistically significant at the 0.05 level. Specifically, the difference in clicks per session yields a p-value of 0.343, while total time spent has a p-value of 0.529. The difference in CTR also remains non-significant ($p = 0.256$).

For binary outcomes, the task completion rate difference is not statistically significant ($p = 0.432$). The bounce rate shows a larger difference between variants, with a p-value of 0.085, which is closer to significance but still above the 0.05 threshold.

Overall, although the right-panel layout consistently shows better performance across multiple metrics, the statistical tests do not provide strong evidence to conclude that these differences are statistically significant.

5. INTERPRETATION

The results indicate that the right-panel layout shows consistent directional improvements across multiple engagement and performance metrics compared to the left-panel layout.

Although the differences are not statistically significant, the observed patterns suggest a potentially meaningful effect. In particular, the lower bounce rate and higher task completion rate imply that users may be more likely to engage with the application and successfully complete tasks under the right-panel layout.

However, the lack of statistical significance indicates that the current evidence is insufficient to support a definitive conclusion. This is likely due to the limited sample size, which reduces the statistical power of the tests.

Therefore, these findings should be interpreted as exploratory rather than confirmatory, highlighting a potential advantage of the right-panel layout that warrants further investigation.

6. CHALLENGES AND LIMITATIONS

This study has several limitations. First, the sample size is relatively small, which reduces the statistical power of the analysis. As a result, it is difficult to detect statistically significant differences even when consistent trends are observed.

Second, the data were generated in a simulated environment rather than collected from real users. Although the dataset was designed to mimic realistic behavior, it may not fully capture real-world user interactions.

Finally, the participant pool is limited and does not represent a broad population, which may affect the generalizability of the findings.

7. CONCLUSION

In conclusion, this A/B test suggests that the right-panel layout is associated with consistent directional improvements in user engagement and task performance compared to the left-panel layout.

However, the current results do not provide statistically significant evidence to support a definitive advantage. Therefore, these findings should be interpreted as exploratory rather than conclusive.

Future work with larger sample sizes and real user data is necessary to validate these observed trends and establish statistical significance.

8. GITHUB LINK

https://github.com/00Kisumi00/STAT5243-Project-3

9. CONTRIBUTION

Yijing Zhang and Yaxuan Hu conducted the data analysis and wrote the main body of the report. Kevin Ma and Haoyun Tong designed the A/B test and developed an alternative version of the application to collect user interaction data. All team members contributed to the experimental design and interpretation of the results.